PS 3

ARE 212, Spring 2026

David Cai, Miriam Gold, Ella Moxley, Geoffrey Yip

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```r
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   4.0.2     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(haven)

path <- "~/projects/ARE212_2026/"
path_ps3 <- file.path(path, "ps3")
path_data <- file.path(path_ps3, "data")

ggplot2::theme_set(
    theme_bw() +
        theme(
            text = element_text(family = "sans"),
            strip.background = element_blank()

        )
    )
```

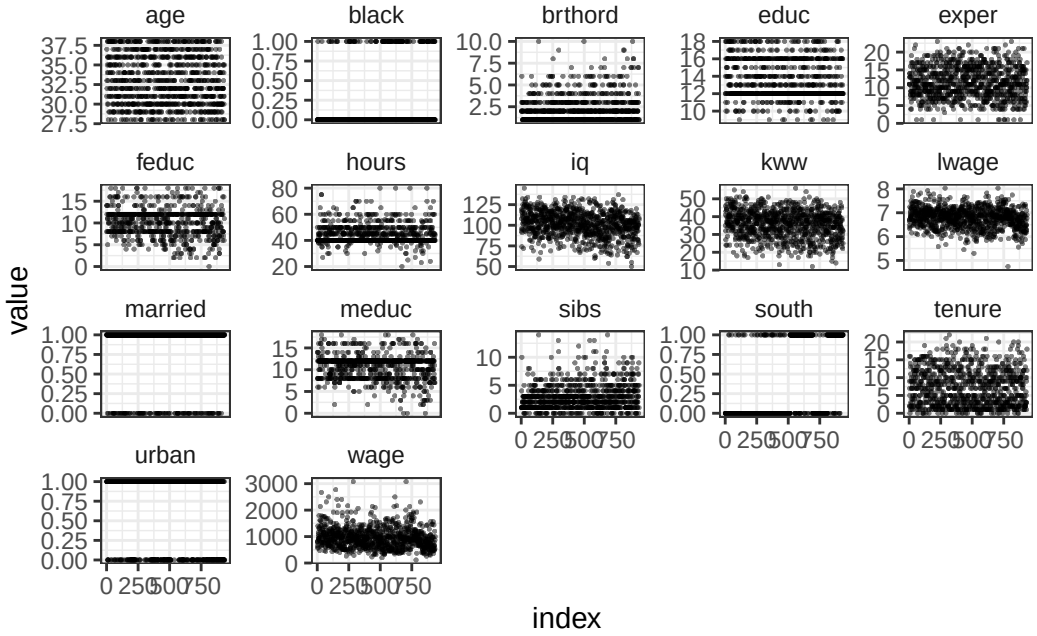
Q1

Read data in and plot series to check data quality.

```
data_nls80 <- read_dta(file.path(path_data, "NLS80.dta"))
```

```
data_nls80 %>%
  mutate(index = row_number()) %>%
  pivot_longer(
    cols = !index,
    names_to = "var"
  ) %>%
  ggplot(aes(index, value)) +
  geom_point(
    shape = 21,
    alpha = 0.5,
    fill = "black",
    stroke = 0,
    size = 0.75
  ) +
  facet_wrap(~var, scales = "free_y")
```

```
Warning: Removed 355 rows containing missing values or values outside the scale range
(`geom_point()`).
```


age
black
brthord
educ
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feduc
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iq
kww
lwage
married
meduc
sibs
south
tenure
urban
wage
value
index

Q2

Q3

Q4

Q5